

Machine Tool Lab

Experiment – 2

Object:- To study the tool makers microscope and checking the alignments of screw thread and cutting tool (five elements of workpiece are checked, outer dia, root dia, pitch, depth, thread angel) (measuring angles end cutting edge angle, side cutting edge angle, lip angle, top back rake angle, clearance angle)

Apparatus:- Tool makers microscope, 6' bolt, cutting tool.

Theory:- A toolmaker microscope is a high-precision optical measuring device used to accurately measure and inspect small tools and mechanical parts. It is a multipurpose instrument that can measure linear distances, angles, and profiles of objects with great accuracy, often up to 1/100th of a millimeter. It is a key tool in metrology and quality control for verifying that parts meet size and quality standards.

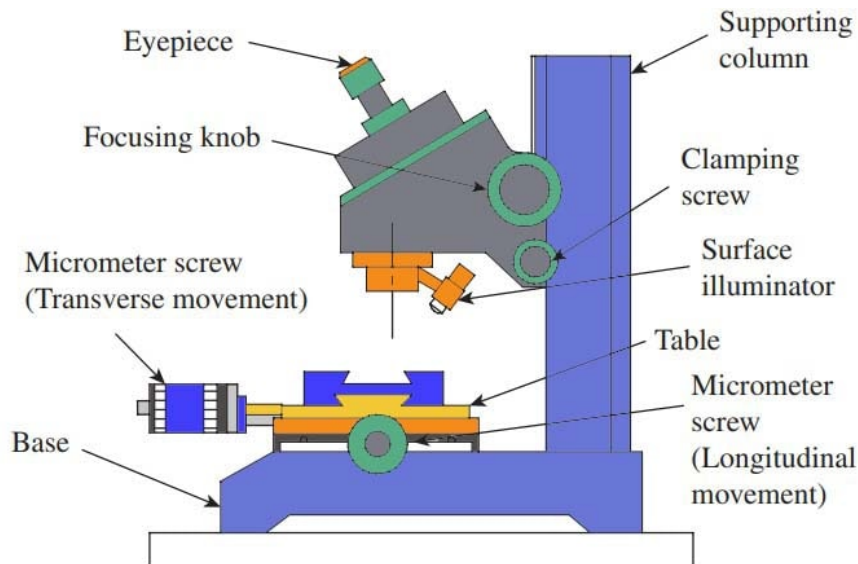


Fig: Tool Maker's Microscope

Applications:-

A tool maker's microscope is primarily used in precision engineering, manufacturing, and quality control for highly accurate dimensional and angular measurements of small, intricate parts. It is suitable for applications in various fields, including:

1. Manufacturing and Engineering: Indispensable for inspecting and measuring small, intricate components like gears, screws, molds, and machine parts.
2. Tool and Die Making: Used for the precise measurement and inspection of dies, molds, punches, and other tools to ensure they meet exact specifications.
3. Electronics and Semiconductor Industry: Utilized for examining and measuring small electronic components, circuit boards, and solder joints, and for failure analysis.

4. **Aerospace and Automotive:** Plays a crucial role in the inspection and dimensional analysis of critical components, such as engine parts and turbine blades, to ensure reliability and safety.
5. **Metrology and Inspection Labs:** Used in laboratories and quality control departments for verifying the accuracy of various gauges, templates, and other measuring tools.
6. **Watchmaking and Optical Industries:** Helps in the precise examination and alignment of small, delicate components like lenses and watch mechanisms.
7. **Research and Development:** Scientists and researchers use it for material analysis, microstructure examination, and studying the properties of materials at a microscopic level.
8. **Jewelry and Gemology:** Aids in the examination and grading of diamonds, gemstones, and precious metals by inspecting their clarity, color, and cut.

A toolmaker microscope's primary function is to make highly accurate linear and angular measurements on small mechanical and electronic parts, such as tools, gears, and screws. It combines the magnification of a microscope with the precision of a measuring instrument to determine the dimensions of a workpiece, including thread pitch, thread angle, and various external and internal diameters. It is used for quality control and metrology in manufacturing settings.

Key functions:-

- a) **Measurement of linear and angular dimensions:** It precisely measures lengths, angles, and diameters of small components.
- b) **Thread and pitch analysis:** It is used to measure the pitch and angle of screw threads, as well as other thread characteristics.
- c) **Comparison and inspection:** It can compare a physical object to a blueprint or drawing to ensure it meets specifications.
- d) **Tool and die inspection:** It inspects the profiles of tools, dies, and cutters, checking for wear and accuracy.
- e) **Magnification and viewing:** It provides a magnified view of small parts, allowing for detailed observation of features and flaws.

Specifications:-

A toolmaker microscope's specifications include magnification (often 30X standard), a measuring stage with travel of up to 50 mm to 100 mm in the X/Y directions, and resolution down to 0.001 mm or 1 μm. Key features also involve two types of illumination (transmitted and oblique), a graduated rotary stage, and various reticles for angular and linear measurements.

1. Optical and measurement specifications

Magnification: Typically 30X, with optional magnifications like 15X, 20X or up to 100X.

Eyepieces: Standard options include 15X or 10X wide-field eyepieces, often with a crosshair reticle.

Resolution: Can be as fine as 0.001mm or 1μm

Linear Accuracy: Dependent on the model, can be around $3 + L/200\mu m$

Repeatability: Approximately $\pm 2\mu m$

Working distance: Varies by magnification, with a common range of 60 mm to 67 mm

2. Mechanical specifications

X/Y Stage travel: Standard travel is often 50 mm × 50 mm with larger options available up to 150 mm × 150 mm

Stage: Features ball-bearing guides for smooth and accurate movement.

Rotary Stage: A circular stage that rotates 360° with a vernier for angle measurement, sometimes with a locking mechanism.

Micrometers: Used for measuring travel, with a least count (L.C.) of 0.01 mm

3. Illumination

Transmitted Light: A sub-stage lamp provides collimated, green-filtered light for viewing contours and transparent objects from the bottom.

Oblique/Incident Light: An adjustable, inclined light source (often twin spot lights) illuminates the surface of opaque objects.

Illumination Control: Separate controls are often provided for each type of illumination to adjust intensity.

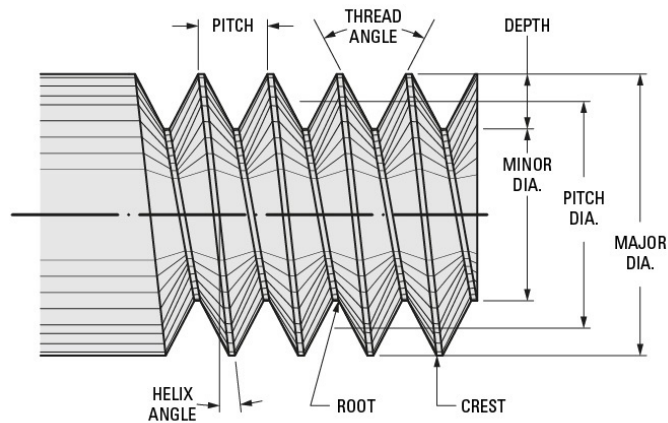
4. Accessories

Reticles: A variety of reticles are available, including those for graph patterns, concentric circles, and specific thread angles.

Attachments: Can include a depth-measuring attachment, center holding device, or V-blocks.

Operating process:-

To operate a toolmaker's microscope, place the object on the stage, turn on the illuminator, and focus the image. Align the object's edge with the reticle's reference hairline, note the micrometer readings, and then move the stage to the next measurement point. The difference between the initial and final micrometer readings gives the linear measurement. For angular measurements, align the reticle's hairline with one side of the angle, take a reading, and then rotate the microscope to the other side and take a second reading; the difference is the angle.



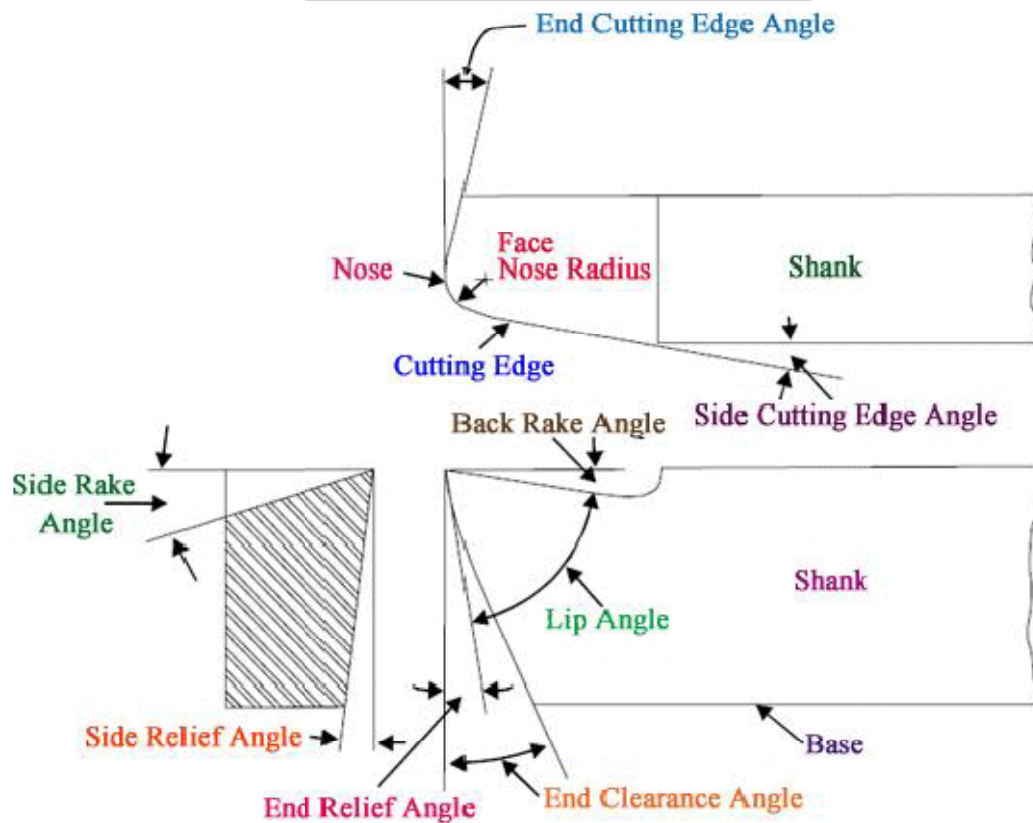
Procedure to measure pitch of a thread:-

1. Turn on the projection lamp and familiarize yourself with the microscope's least count, linear and angular readings, and the specimen's features.
2. Place the thread specimen on the glass table and use the stage controls to move it so the thread pitch is parallel to the X-direction movement.
3. Use the X-direction micrometer and the microscope's controls to align the intersection of the crosshairs with the crest of the first thread.
4. Note the micrometer reading for the X-axis.
5. Rotate the X-direction micrometer to move the specimen until the crosshairs align with the crest of the next successive thread.
6. Note the new micrometer reading.

Procedure to measure the angle of a thread (e.g., the included angle)

1. Rotate the stage and eyepiece until one of the crosshairs coincides with one flank of the thread.
2. Note the initial reading on the angular scale or protractor.
3. Rotate the stage or the screen until the same crosshair coincides with the other flank of the same thread.
4. Note the final angular reading.
5. The included angle is the difference between the two angular readings.

Single Point Cutting tool Diagram



Procedure to measure end clearance angle of cutting tool

1. Align the horizontal crosshair with the top edge of the tool shank and set the angle dial to zero.
2. Rotate the tool bit (or the stage) so the cutting edge is aligned with the horizontal crosshair.
3. Rotate the angle dial counter-clockwise until the vertical crosshair aligns with the cutting edge, then note the reading from the angle dial.
4. Repeat for other angles like side rake and side clearance by changing the tool's orientation on the stage and repeating the alignment procedure.

Procedure to measure linear dimensions (like tool nose radius or flank wear)

1. Align the vertical or horizontal crosshair with the starting point of the feature you want to measure.
2. Move the X and/or Y digital micrometer to bring the crosshair to the endpoint of the feature.
3. The difference between the initial and final readings on the micrometer gives the linear measurement. For example, to measure flank wear, you would align the crosshair with one side of the wear land and then with the other, taking the distance between the two readings.

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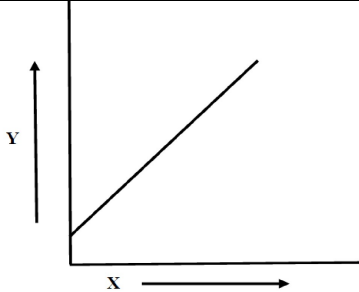
Experiment – 3

Object: To study the pneumatic comparator, calibration and find out the inclination of a given piece.

Equipment: Solex type pneumatic comparator and compressor

Calibration the Pneumatic Comparator: Calibration of pneumatic comparator is down because it helps and guides in taking the reading during regular use and we are also able to understand the taking the initial reading (range) of the system.

Table: Reading for Calibrations

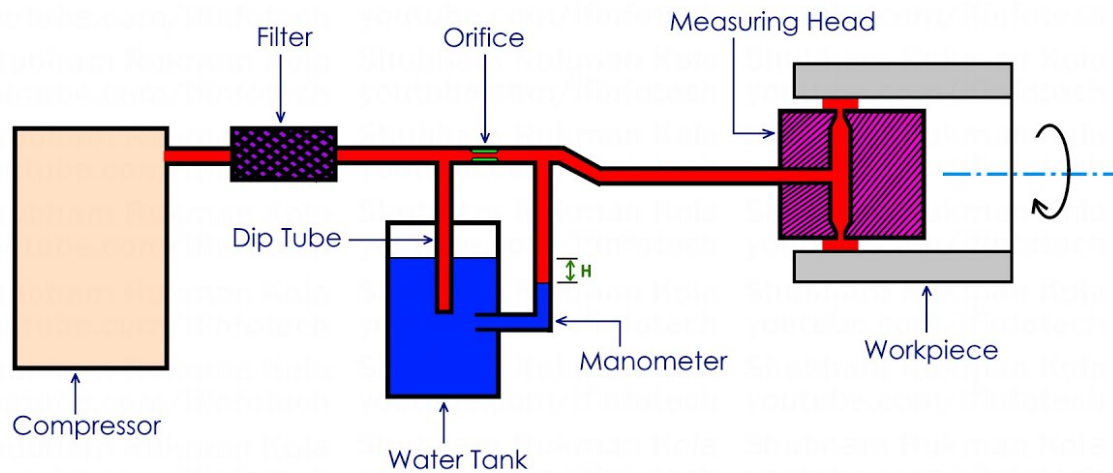
S. No.	Dial Gauge Reading (X)	Manometer Reading (Y)	Graph Characteristic curve S v/s Y
1.	0.0		
2.	0.1		
3.	0.2		
4.	0.3		
5.	0.4		
6.	0.5		
7.	0.6		
8.	0.7		

Method of Calibration: By starting the compressor and supplying compressed air at constant pressure. This air passes through the controlling and measuring orifices. Start taking readings from the manometer by using dial gauge for the required gap between the measuring orifice and specimen. Note the initial reading when the backpressure start and level of water in manometer tube starts falling down. Note the readings using dial gauge in 0.1 mm reducing gap between measuring orifice and specimen up to the surface of the measuring orifice. The table and graph are plotted as required.

Pneumatic comparator:

The pneumatic back pressure sensor identifies the object and provides a pressure signal. The intensity of back pressure signal from the sensor depends upon the relative position of the object from the sensor provided supply pressure to the sensor is maintained constant. Hence, for a slight change in the distance of the object of the sensor ranges in microns gives a considerable change in back pressure and thus sensor becomes indispensable in the field of repetitive gauging to close tolerance. The instrument makes use of a constant supply of air at controlled pressure. The gauging element is based upon the principle of allowing elative small quantity of air to escape between the outside or inside of the gauging element and the work.

Solex Pneumatic Comparator



The velocity or pressure variation caused by the escaped air is then measured on a sensitive amplifying device. In most application the pressure variation are measured and the pneumatic comparator unit is calibrated against master gauge so that the measurement obtained with the gauging head represents deviation from the master gauge dimension thus the pneumatic gauge is basically compared at measuring difference and not basic or standard dimensions.

Operating principle:

Back pressure sensor is most commonly used position sensor. The principle of pneumatic nozzle and taper device is utilized. The sensor consists of a constant pressure supply separated from back pressure chamber by a fixed restriction with form or control orifice. The sensor is shown figure when the distance of the object changes with nozzle =, back pressure that is the pressure reflected on upstream side of a system due to downstream operation, becomes the function of displacement of the object, provided supply pressure remains constant. The backpressure characteristics i.e., graph between back pressure/supply pressure v/s non dimensional displacements is to be drawn. It is seen that a portion of curve which is approximately leaner can be used for a sensing system. For digital system application the extreme point of the characteristics is only points of interest so that a pressure is attained above and below the switching pressure of device has wider application in air gauging to closed tolerance and in proximity sensing. It is claimed that by using backpressure sensor under controlled condition a precession gauging of a few microns can be obtained.

Study of solex pneumatic comparator

The principle of this type of measuring instrument depends upon the air flow between the gauging head and hole in which it is inserted. Main parts of the solex pneumatic comparator are as below:

1. Threaded spindle

2. Knurled nuts
3. Socket
4. Measuring orifice
5. Controlled orifice
6. Water tank chamber
7. Air chamber
8. Wooden box
9. Primemover (compressor)

Referring to figure, the air under constant pressure obtained A dip tube controlling device passes through a control jet **B** which meters the amount of air delivered to the gauge. At **C** there is flexible air tight tube between **B** and the gauging head jet **G**. A manometer or air pressure gauge is connected to the branch pipe at **D**. The intermediate chamber between **B** and **G** is indicated as **F**, the work to be measured at **I** and the instrument stand at **E**.

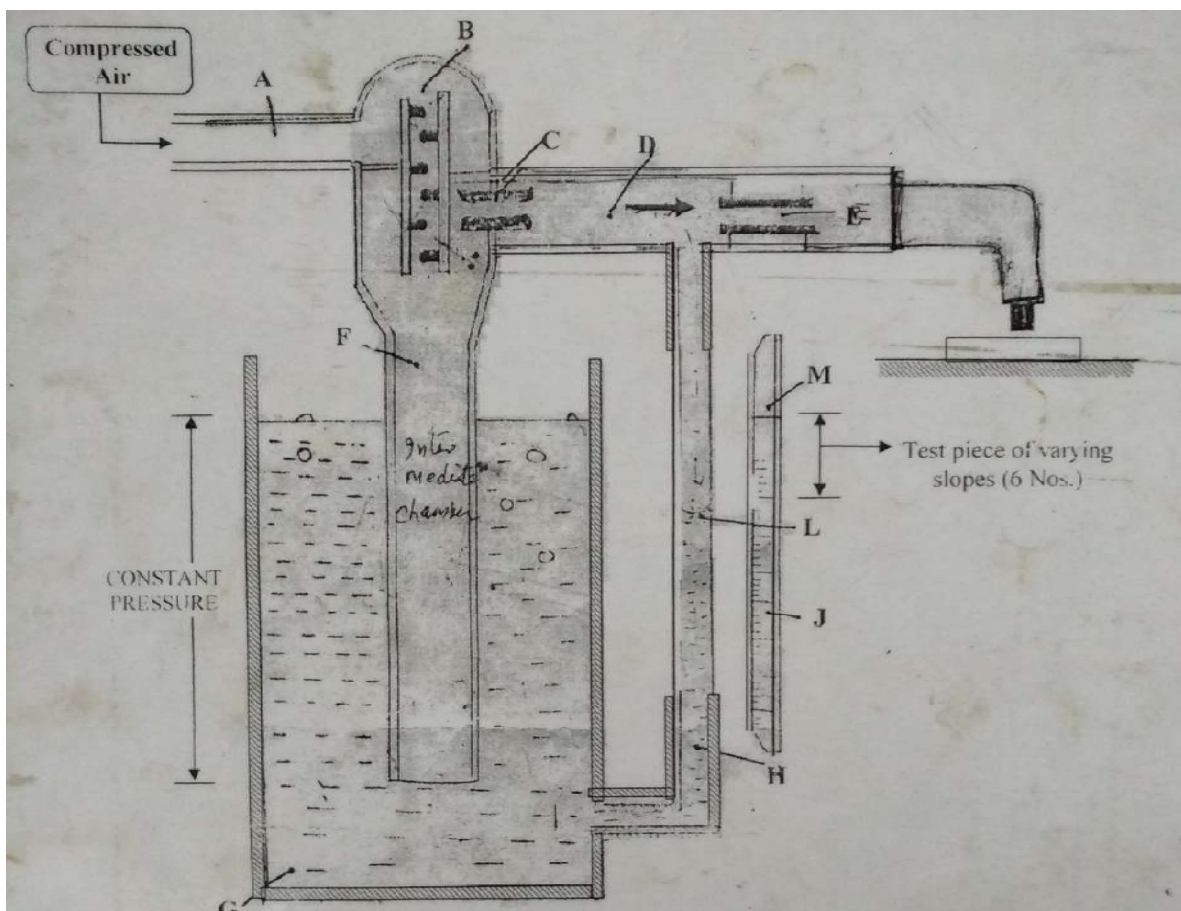


Fig: Air pressure regulator

Air under pressure from a main supply is set at a constant pressure and then flows through the control jet **B** to the orifice known as the gauging head at **G**. According to the amount of air escaping through the latter jet a pressure is built up to definite value between **B** and **G**, so that the flow through them is exactly balanced and independent of outside conditions. The pressure depends only upon the relative value of the orifices and the control jet **B** is constant, the pressure then depends only upon the second orifice **E** i.e. the clearance between the gauging head and

the work which is the example shown in figure is a flat surface parallel to the lower surface of the gauging head. The manometer measures this pressure which is then taken as a measure of the clearance between the gauging head and the work. The air pressure regulator works on the dip tube principal shown at **A** is the connection to main air supply at **B** certain restrictions jets, to limit the maximum amount of air delivered by air line the control jet is shown at **C** the intermediate chamber at flexible connection to the gauging head **G** dip tube at **F** water container at **G** manometer at **J** and glass tube of manometer at **H** with water level at **I**.

In use air from the main supply at a restricted in amount by **B** passes down the dip tube **F** and just bubbles out, this maintaining constant pressure in dip tube. The actual value of the pressure is that of the head of water above the bottom and of the dip tube. When the gauging head in operation the pressure in the intermediate chamber will be less than that in the dip tube its value being shown **B**, The vertical distance between the water level **M** and lower part scale **L** of the water in the manometer tube. The scale of the letter is graduated according to the dimensions of part to be measured and each gauge has its own particular scale.